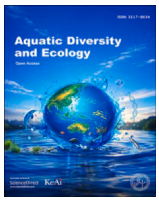
Aquatic Diversity and Ecology 1 (2026) 100009

Contents lists available at ScienceDirect

Aquatic Diversity and Ecology

journal homepage: www.keaipublishing.com/ade/



Research Article

# YZFishDB: A curated DNA reference database for the fishes of the Yangtze River Basin

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ARTICLE INFO

*Keywords:*
Biodiversity
Environmental DNA
Freshwater ecology
Fish ecology
Metabarcoding
Actinopterygii

ABSTRACT

The utility and precision of DNA metabarcoding are heavily dependent on the quality of the corresponding reference database, especially in understudied ecosystems. Public databases are often relied upon, especially for studies in species-rich ecosystems, despite potential issues of accuracy and reliability. We developed a curated DNA reference database (YZFishDB) by combining four public reference databases and representative sequences of underrepresented taxa for ray-finned fishes of the Yangtze River Basin (YRB), to improve taxonomic identification, particularly for environmental DNA (eDNA) metabarcoding. We compared taxonomic assignment results for a set of eDNA data from the YRB using YZFishDB and the MitoFish reference database to test their utility. YZFishDB included 54,546 unique sequences covering 503 taxa. The taxonomic assignment comparison showed that both reference databases (YZFishDB and MitoFish) identified 118 shared taxa, while YZFishDB identified 16 additional taxa and MitoFish identified 7 additional taxa. All ASVs assigned to unique taxa by MitoFish were assigned to other taxa by YZFishDB, with better BLAST quality metrics. The development of YZFishDB provides a valuable resource towards improving DNA metabarcoding results as eDNA surveys continue to increase in popularity in the YRB.

## 1. Introduction

DNA metabarcoding has rapidly become a cornerstone of biodiversity assessment and species identification, driven by advances in sequencing technologies and declining costs (Morris et al., 2024; Yao et al., 2022). In ecological studies, DNA metabarcoding increasingly replaces traditional morphological identification, particularly for challenging cases such as cryptic species detection or sampling constraints (Beng and Corlett, 2020; Bessey et al., 2020; Yao et al., 2022). In addition, DNA-based methods allow for more detailed delineation of species movement and population exchanges, which has greatly benefited research in population dynamics and biodiversity assessments (Adams et al., 2019; Beng and Corlett, 2020; Verhelst et al., 2022). The growing popularity of DNA metabarcoding has driven standardization and improved data quality for both specimen-derived and environmental DNA (eDNA) samples (Burian et al., 2021; Loeza-Quintana et al., 2020; Pereira et al., 2021). The combination of higher data quality and lower costs has allowed researchers to broaden their study scope, especially in previously understudied regions and communities (Campbell et al., 2022; Manandhar et al., 2023; Sanchez et al., 2025).

The accuracy of DNA metabarcoding hinges on the quality of reference databases, which link sequences to taxonomic identities (Bessey et al., 2020; Ficetola et al., 2016; Magoga et al., 2022; McClenaghan et al., 2020). Ideally, such databases should comprehensively represent the regional species pool and contain complete, high-quality sequences for relevant genetic markers. However, assembling exhaustive local databases is often impractical for species-rich ecosystems. Therefore, public DNA reference databases are often relied upon for taxonomic

Peer review under the responsibility of Editorial Office of Aquatic Diversity and Ecology.
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https://doi.org/10.1016/j.ade.2026.100009
Received 27 April 2026; Received in revised form 29 May 2026; Accepted 4 June 2026
Available online 10 June 2026
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assignments when studying large ecosystems, despite potential issues, particularly regarding their accuracy and reliability (Chorlton, 2024).

Despite their utility, public repositories suffer from issues including misidentifications, contamination, and inconsistent metadata, all of which can compromise species assignment accuracy (Bagheri et al., 2020; Keck et al., 2023; Magoga et al., 2022; Sangster and Luksenburg, 2021). These challenges are especially pronounced in biodiversity hotspots such as the Yangtze River Basin (YRB), where high species richness and endemism demand comprehensive, rigorously validated sequence coverage. Moreover, false positives, especially for non-native, rare, or endangered taxa, pose significant concerns when metabarcoding results are not cross-checked against verified species occurrence records.

Since 2021, a ten-year basin-wide fishing ban has been enacted in the YRB to prevent further declines in fish stocks after decades of overexploitation (Mei et al., 2020). While fisheries assessments comparing pre- and post-fishing ban have shown positive trends towards ecosystem recovery, the community dynamics and ecosystem stability remains uncertain as studies have observed signs of imbalance (Du et al., 2025). Under the reduced access for fisheries surveys in the YRB following the implementation of the fishing ban, there has been a boom in eDNA surveys and studies in the basin (See Li et al., 2025; Lyu et al., 2024; Qian et al., 2023; Shen et al., 2024), with increasing efforts being invested in improving eDNA survey quality and reliability in freshwater ecosystems (Hwang et al., 2025).

However, access to a high-quality DNA reference database is a major hurdle for the YRB as a whole, especially given the resources required to collect and build a comprehensive database for such a large ecosystem. While recent studies have begun to fill in this major data gap, the accessibility and coverage of these reference databases remain limited. To date, we found only two local reference databases pertaining to the YRB: (1) a basin-wide COI reference database covering 123 species (available on Barcode of Life Datasystem (BOLD); Shen et al., 2019) and (2) a regional multimarker (12S, 16S, COI) reference database for the Three Gorges Reservoir covering 120 species (no public repository; Xie et al., 2026).

As such, eDNA studies in the YRB still largely relies on public reference databases, frequently using a combination of National Center for Biotechnology Information (NCBI) nuccore, MitoFish or BOLD (Cheng et al., 2023; Feng et al., 2025; Jia et al., 2020; Li et al., 2022; Peng et al., 2026; Shen et al., 2023, 2016). Typically, additional post hoc processing of taxonomic assignment results is required to filter out spurious taxa, not to mention the aforementioned potential pitfalls of public repositories, including sequence mislabeling and a lack of data for rare and uncommon taxa. In this context, there is a need for a curated DNA reference database for the fishes of the YRB to improve the quality and reliability of eDNA metabarcoding in this major ecosystem.

To address these gaps, we compiled a comprehensive list of fishes in the YRB, including known and potentially introduced non-native fishes, from primary and secondary sources and developed "YZFishDB", a curated reference database tailored for the molecular identification of ray-finned fishes (Actinopterygii) in the YRB (Fig. 1). By integrating and validating sequences from commonly used public repositories (i.e., BOLD, MIDORI2, MitoFish, and NCBI) and additional sequences for rare taxa, with a focus on regionally relevant markers (e.g., 12S and COI), this database would provide users with a tool that balances comprehensive coverage with better taxonomic reliability which can support combinations of taxonomic identification methods.

## 2. Methods

### 2.1. Fish list of the Yangtze River Basin

We compiled the most up to date list of the fish species found in the Yangtze River Basin (YRB) based on four main sources, i.e., "cnfishbase" (Lu et al., 2023), "Changjiang Fishes" (Institute of Hydrobiology, 1976),

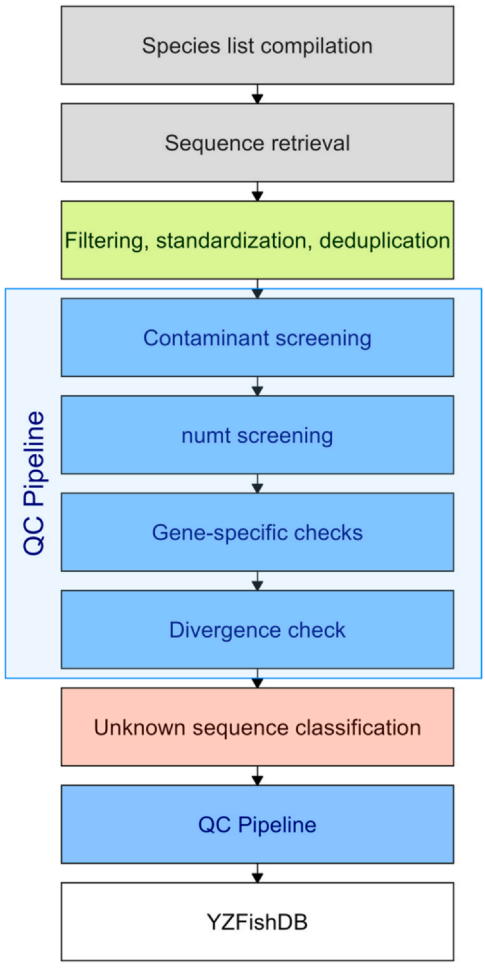



**Fig. 1.** The workflow describing the development of the local DNA reference database, YZFishDB, from data compilation to curation.

and "Distribution, endemism, conservation status of fishes in the Yangtze River Basin, China" (Ye et al., 2011), and "Status of aquatic organisms resources and their environments in Yangtze River system (2017–2021)" (Yang et al., 2024). We further supplemented the list for non-native species with "Non-native freshwater fish species in China" (Xiong et al., 2015) and "Status and ecological risks assessment of alien fish in the Yangtze River Basin" (Wan et al., 2025).

Despite its early publication date, "Changjiang Fishes" (Institute of Hydrobiology, 1976) is still considered the foundational source for the freshwater fishes of the Yangtze River, from which most YRB freshwater fish species lists are derived. Nevertheless, Ye et al. (2011) is the more commonly cited source in recent decades for describing the total number of fish species found in the YRB and can be considered the currently well-accepted species list. Yang et al. (2024) present more recent results from survey efforts in the YRB, with additional information on several introduced species that are becoming increasingly prevalent.

We built upon the general framework of cnfishbase to cross-reference the species list from all four sources in chronological order. Due to taxonomic updates over the past several decades, we primarily referenced Eschmeyer's Catalog of Fishes (Fricke et al., 2025). We adopted the currently valid names as our standardized taxonomic names, where applicable. Specifically, we cross-checked each unique fish species name in our sources against Eschmeyer's Fish Catalog (Fricke et al., 2025) to confirm its validity. When a species name from our sources was not listed as an accepted valid name, it was standardized to the currently listed valid name. In cases where the scientific names were absent from the Eschmeyer's Catalog of Fishes, we cross-

referenced its listed common names across our references, supplemented by FishBase, to identify them. Major taxonomic changes, e.g., lumps and splits, were only accepted after confirmation based on the corresponding literature.

Additionally, we also retained all outdated taxonomic names. We recorded the listed synonyms from Eschmeyer's Catalog of Fishes, along with the Fishbase species code in a lookup table to facilitate subsequent searches and updates. To facilitate the identification of introduced species, we recorded the biostatus ("native" or "introduced") of the species in the YRB. Information for introduced species were compiled from Lu et al. (2023), Ye et al. (2011), Yang et al. (2024), Xiong et al. (2015), and Wan et al. (2025), with "Changjiang Fishes" (Institute of Hydrobiology, 1976) only recording native fishes.

### 2.2. Data mining and compilation

We included four DNA sequence databases to construct our YRB fish mitochondrial DNA (mtDNA) reference database, i.e., BOLD (Ratnasingham et al., 2024), MIDORI2 (Leray et al., 2022), MitoFish (Iwasaki et al., 2013), and NCBI nucleotide "nuccore" database (Sayers et al., 2025), with slight differences in the data compilation process for each database. For the BOLD database, we retrieved all sequences attributed to the class Actinopterygii from its public database on 11 April 2025; from the MIDORI2 database, we only retrieved data from the "longest" and "uniq" srRNA (12S) sequences from GenBank 263 (13 October 2024); for the MitoFish database, we retrieved the entire mitogenome database on 26 February 2025 because MitoFish is one of the few reference databases that is specific to fishes. We then subsequently extracted the sequences attributed to taxa found on our YRB fish species list from the retrieved sequence data from these three sources (i.e., BOLD, MIDORI2, and MitoFish) and standardized the format to include data origin, sequence gene (12S, 16S, COI, complete genome, or others), and sequence completeness (complete or partial).

Due to the size and variety found on the NCBI database, we opted to specify our search term to cover based on our fish species list while still covering a relatively broad scope (Species represents the iterative species name from the species list):

> *"Species"[Organism] AND ("mitochondrion"[Filter] OR "mitochondrial"[Title] OR "mtDNA"[Title] OR "mitogenome"[Title]) AND (("MT-RNR1"[Gene Name] OR "MTRNR1"[Gene Name] OR "12S"[Title] OR "12S rRNA"[Title]) OR ("MT-RNR2"[Gene Name] OR "MTRNR2"[Gene Name] OR "16S"[Title] OR "16S rRNA"[Title]) OR ("MT-CO1"[Gene Name] OR "COI"[Title] OR "CO1"[Title] OR "COX1"[Title] OR "cytochrome c oxidase"[Title]) OR ("complete genome"[Title] OR "mitochondrial genome"[Title]) NOT ("unverified"[Title] OR "environmental sample"[Title] OR "uncultured"[Title] OR "MUGB"[All Fields] OR "cDNA"[All Fields]) NOT ("synthetic construct"[Organism] OR "vector"[Title])*

We searched and retrieved only mtDNA data which included names of the specified species with preferences for 12S, COI, complete genomes within the NCBI nucleotide ("nuccore") database on 26 June 2025 using a customized R script which leveraged functions from the "rentrez" and "tidyverse" R packages (Wickham et al., 2019; Winter, 2017). The resulting sequence data was manually cleaned and processed to remove incorrectly attributed sequences, and was subsequently standardized similarly with sequence data from the other sources. A SQLite database was set up to archive sequence data from all four sources, separated by database origin, using the "RSQLite" R package (Müller et al., 2025).

The compiled database was then supplemented with select (COI, 12S, and 16S) sequences obtained from verified field specimens from the YRB (Table S1). These sequences covered 14 taxa, with little or no relevant sequences available in public repositories, representing rare taxa from the headwaters and upper reaches of the YRB. Extracted muscle tissues from each specimen were preserved in 100% ethanol at 4 °C prior to DNA extraction. High molecular weight DNA was extracted using QIAGEN MagAttract HMW DNA extraction kit (Maryland, USA). DNA integrity was assessed by 1% agarose gel electrophoresis, and the purity and concentration of DNA were measured using a NanoDrop One spectrophotometer (Thermo Fisher Scientific, USA) and an Invitrogen Qubit fluorometer (Thermo Fisher Scientific, USA), respectively. A short-read library with an average insert fragment length of approximately 350 bp was constructed using the MGIEasy Universal DNA Library Preparation Kit (MGI, China) with the PE159 (150 bp both ends) sequencing method. The raw reads obtained from sequencing were subjected to quality control and filtering using FastP (v.0.23.4) to obtain high-quality clean read. Mitochondrial genomes from all samples were successfully assembled into complete circular structures using de novo assembly of the mitochondrial genome with GetOrganelle (v1.7.5), which was then validated through sequence alignment. The assembled mitochondrial genomes were annotated following the annotation procedure from the MitoFish database (https://mitofish.aori.u-tokyo.ac.jp/).

### 2.3. Data processing and quality control

All data processing steps were performed using ad hoc R scripts and bash scripts, unless otherwise specified (R Core Team, 2025). The four standardized sequence datasets (i.e., BOLD, MIDORI2, MitoFish, and NCBI) were merged based on species names and sequences to compile a sequence reference database. To ensure that only a one-to-many relationship exists between species names and sequences, duplicates based on species name and sequence were first removed, leaving unique species-name and sequence combinations. Next, ambiguous sequences attributed to multiple species (many-to-many or many-to-one) were then extracted for manual processing before recombining. These ambiguous sequences were either removed or merged depending on the combination of species attributed to them. For example, sequences that were attributed to different variants or subspecies were merged at the species-level resolution. Similarly, sequences that were attributed to the same species under different names were also merged under the valid species name according to the Eschmeyer's Catalog of Fishes (Fricke et al., 2025). In cases where the same sequence was attributed to distinct species, we manually compared their source metadata to verify the identity of these entries. If the source metadata does not resolve the ambiguity, the species names were merged when one of the species had fewer than 5 entries; otherwise, these ambiguous sequences were removed.

#### 2.3.1. QC pipeline

We then conducted contaminant screening against non-Actinopterygii sequences using selected sequences from *Homo sapiens*, *Escherichia coli*, and *Pseudomonas* spp. In addition, we screened for potential nuclear mitochondrial DNA segments (numts) using the nuclear genomes of *Danio rerio*, *Cyprinus carpio*, and *Carassius auratus* (Antunes and Ramos, 2005). Numts are mtDNA copies that are transferred to the nuclear genome during evolution and are often misannotated due to their similarity to mtDNA segments, which can lead to false positives for additional variants due to higher mutation accumulation (Uvizl et al., 2024). The contaminant screening for both non-Actinopterygii and numts were conducted using BLASTN locally where entries matching the contaminant sequences where e-value $\leq 1 \times 10^{-20}$ and pident $>$ 95% were flagged, verified, and removed. We selected these stringent quality metrics to minimize false positives (sequences wrongly identified as contaminants), despite the trade-off of potentially retaining false negatives to maximize sequence retention in the database. A gene-specific check was conducted for sequences labeled "12S", "16S", or "COI". COI sequences were checked for internal stop codons and potential frameshifts based on codon alignment using the multiple alignment by the fast Fourier transform (MAFFT) and flagged accordingly. 12S and 16S sequences were checked and flagged if the total

sequence length was less than 100 bp ("short") and if the sequences contain gaps ("gaps"). For sequences of any gene type for any species with more than 5 sequences, potential mislabeled species or hybrids were identified by comparing the median length of branches based on NJ trees (FastTree) constructed from aligned (MAFFT) conspecific sequences, where sequences that differed more than 5 times the median or 0.02 (2%) substitutions per site (whichever larger) were flagged as potentially erroneous ("divergent"). In consideration of known intraspecific variations between different genes and species, we opted for this adaptive threshold (Collins and Cruickshank, 2013) which combined (1) a distribution-based outlier rule based on the median (Comte et al., 2023) and (2) an absolute floor aligned with common intraspecific divergence benchmark for vertebrate COI (2%; Hebert et al., 2003; Ward et al., 2005). By selecting a conservative relative threshold of 5 times the median tip branch length, this step aims to identify extreme outliers that are most likely divergent sequences. Lastly, sequences flagged as "divergent", "frameshifted", and "contaminant" were verified before removal.

#### 2.3.2. Gene type classifier

This intermediate database was passed through a gene-type classifier trained using a machine-learning approach to classify unlabeled sequences (labeled "other"). Our gene type classifier was based on a random forest implemented with the 'ranger' engine in the 'tidymodels' framework (Kuhn and Wickham, 2020; Wright and Ziegler, 2017). The random forest model was trained on z-score-normalized, zero-variance-filtered feature vectors comprising sequence length, GC content, GC skew, AT content, and AT skew, derived from raw nucleotide string counts. Model hyperparameters were set to 500 decision trees, 3 features considered per split (mtry = 3), and a minimum node size of 5. The training data (80:20 stratified split) consisted of 22,579 sequences (12S: 2008; 16S: 1626; COI: 16,791; genome: 1375; multiple: 779) from the intermediate database, where sequences flagged as "divergent" and "contaminant" were excluded to ensure the model learned from high-confidence references. The classifier model achieved high predictive accuracy ($accuracy = 0.988$, $\kappa = 0.972$), with a class-level performance showing relatively strong recall (macro recall = 0.952; ranging from 0.84 to 1.00) and balanced performance overall ($F_1 = 0.975$; ranging from 0.91 to 1.0) across all sequence types on the 20% hold-out test set (Table S2). Unlabeled sequences greater than 100 bp were passed through the trained gene type classifier, and a 0.8 threshold was used to calculate the maximum class probability. The newly classified sequences were validated by passing them through the QC pipeline again to flag potential false-positive classifications by our machine learning gene type classifier. Consequently, sequences flagged in the second round of QC as "divergent" and "contaminant" were, again, verified before removal.

We conducted a post hoc barcode gap analysis across all target markers (COI, 12S, 16S and genomes) to further evaluate the taxonomic resolution capacity of the reference database. We computed the barcoding gap metrics for all species-marker combinations with 5 or more sequences, after sequence alignment (using the "DECIPHER" R package; Wright, 2024), based on pairwise p-distances calculated from normalized Hamming distances. Specifically, we computed both a median-based and a percentile-based threshold to reduce sensitivity to potential outlier sequences, such that a median barcoding gap was present when the median intraspecific distances was less than the median interspecific distances. At the same time, a percentile barcode gap was present when the 95th percentile of intraspecific distances was less than the 5th percentile of interspecific distances.

The resulting full reference database can then be subsetted based on the required sequence type (e.g., 12S or COI) and further specified for selected primers. Finally, higher taxonomic-level information (from genus to kingdom) for each species was added to this reference database and stored in a SQLite database. The reference database can be exported as a FASTA file with selected formats for four mainstream bioinformatic pipelines ("QIIME2", "USEARCH", "DADA2", "BLASTN") for subsequent use, e.g., taxonomic assignments.

### 2.4. Taxonomic assignment comparison

Raw DNA metabarcoding data ($n = 197$ sites across the YRB, developed by Xiong et al., 2025; Fig. S1) were used to test the efficacy of our curated database. These samples were amplified using the Ac12S fish metabarcoding primer set (Xiong et al., 2025; Zhang et al., 2022) with the bioinformatic analyses were performed following the DADA2 workflow (Callahan et al., 2016) for primer removal, quality filtering, and denoising to obtain the amplicon sequence variants (ASVs).

We compared the ASV taxonomic assignments between YZFishDB, our developed reference database, and a commonly used conventional public database, i.e., MitoFish mitogenome database (retrieved on 26 February 2025). We performed the taxonomic assignment using BLASTN (percent identity $\geq 99\%$, e-value $\leq 1 \times 10^{-20}$). The BLAST results were summarized at the species level, retaining the maximum percent identity (pident), bit-score (bitscore), alignment length (length), subject coverage (scovs), and query coverage (qcovs), and the minimum e-value (evalue), mismatch, and gap opening values (gap). Bitscore per base (bitscore_pb; $\frac{bitscore}{length}$) and the combined coverage (ccovs) was calculated as qcovs corrected by scovs ($ccovs = \sqrt{qcovs \times scovs}$). The summaries were then ranked based on multiple alignment quality metrics in the following order: (1) descending bitscore, (2) descending bitscore_pb, (3) ascending evalue, (4) descending ccovs, (5) descending pident, (6) ascending mismatch, (7) ascending gap, (8) descending species hit counts. The resulting top-ranked hits were retained as the most likely taxonomic assignments. Remaining hits with identical alignment metrics across species were flagged for manual verification.

## 3. Results

### 3.1. Fishes of the Yangtze River Basin

The final list of YRB fishes included 567 unique species (221 genera; 65 families; 28 orders). Of these, 494 (approx. 87.13%) are native species and 74 are non-native species (approx. 13.05%; Table S3). This species list included several more species than previous lists, primarily due to the inclusion of multiple potential non-native species that have yet to be confirmed in the literature for the YRB.

### 3.2. Reference database compilation and quality control

Before any data quality control, we retrieved a total of 57,807 sequences across all five data sources (BOLD: 6423; MIDORI2: 1918; MitoFish: 47,578; NCBI: 1848; and local: 40; Table S1; Table S4). After removing duplicate sequences from the four source databases and our local sequence data, a total of 57,807 unique sequences were compiled prior to quality control screening. The first round of our quality control process retained 57,185 sequences after removing all "contaminant" and "divergent" sequences with confirmed genes (e.g., 12S, 16S, and COI; Fig. 2). The ML classifier identified 21,019 sequences (12S: 148, 16S: 381, and COI: 20,490) from the previously unconfirmed sequences ("other") with high confidence (0.8 probability threshold), leaving 13,587 sequences unconfirmed.

After removing the remaining low-quality sequences (i.e., contaminants, divergent, short sequences, and sequences with gaps) following the second round of quality control, the resulting database included 54,546 unique sequences for 504 unique species (216 genera; 65 families; 29 orders). Of these, YZFishDB included 1913 12S sequences (430 species, 209 genera, 64 families, 28 orders), 1975 16S sequences (241 species, 149 genera, 58 families, 27 orders), 35,583 COI sequences (500 species, 216 genera, 65 families, 30 orders), and 1368 complete

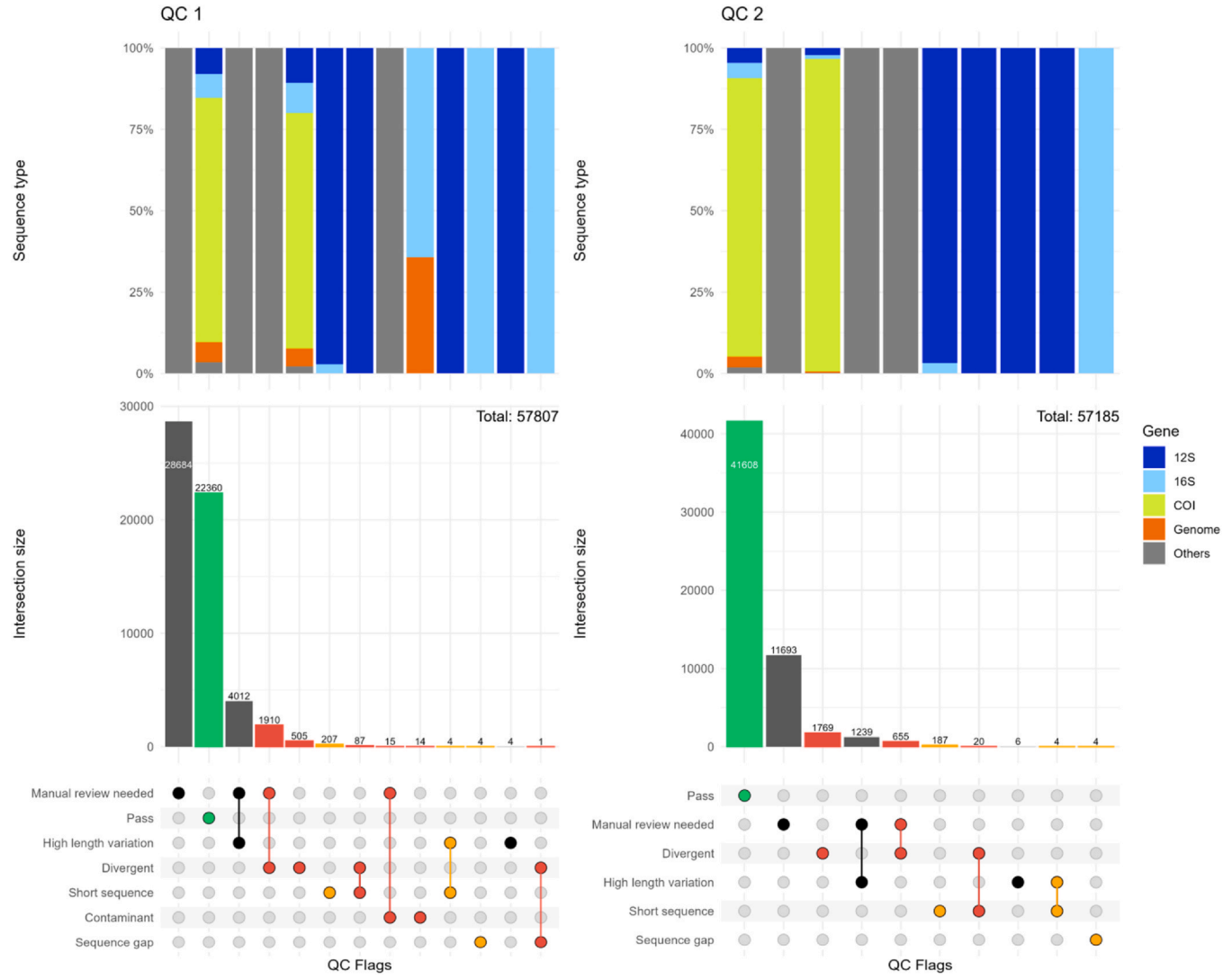



**Fig. 2.** Count of the combinations of QC flags following the first and second round of the QC pipeline and their associated sequence type (12S, 16S, COI, Genome, etc.) proportions. Upset bars are highlighted according based on the relevant QC results: green indicates sequences which passed; red indicates serious error flags (contaminant and divergent); yellow indicates low quality sequences (short sequences, gaps in sequences). All sequences with serious and low quality flags were removed after each round of QC.

genomes (433 species, 205 genera, 62 families, 27 orders), with the remaining sequences being sequences from multiple genes or unlabeled sequences (Fig. 2; Fig. 3; Fig. S2).

When compared to the MitoFish reference database (49,491 sequences), YZFishDB had more total sequences (54,546), with increases across all gene categories, reflecting the limitations of relying on a single reference database (Fig. 4). For the gene sequences most relevant to our taxonomic comparison study, the number of 12S sequences more than doubled (YZFishDB: 1913; MitoFish: 679), while full genome sequences increased by 88 (YZFishDB: 1368; MitoFish: 1280). A taxonomic comparison showed that all species in the MitoFish database remained present in YZFishDB after curation, with YZFishDB containing an additional 47 taxa (Fig. 4).

The post hoc barcode gap analysis suggested that overall, median intraspecific p-distance (0.08; IQR: 0.015–0.108) was less than the median interspecific p-distance (0.107; IQR: 0.097–0.138), with 87.6% of species-marker combinations having a median barcode gap (Table S5). However, percentile barcoding gaps were absent for most species-marker combinations (87.4%), suggesting substantial intraspecific variability within some taxa despite quality filtering (Table S5). Nevertheless, the widespread presence of median barcode gaps suggests that most markers still retain discriminatory power for taxonomic assignment despite overlap at the distribution tails.

### 3.3. Taxonomic assignment comparison

The resulting taxonomic assignment using the YZFishDB identified 1340 ASVs compared to the MitoFish database which identified 1326 ASVs, of which 1303 assignments (95.6%) were common between the two databases, 37 assignments (2.7%) were unique to YZFishDB and 23 (1.7%) were unique to MitoFish (Fig. 5).

Taxonomic assignment results from both YZFishDB and MitoFish included several instances of ASVs matching more than one taxa sharing identical assignment quality metrics, with YZFishDB having seven ASV pairs across six species (*Hypophthalmichthys molitrix* vs. *Hypophthalmichthys nobilis*, *Macrognathus aculeatus* vs. *Sinobdella sinensis*, and *Triplophysa lixianensis* vs. *Triplophysa orientalis*) and MitoFish having 11 pairs across six species (*Megalobrama terminalis* vs. *Parabramis pekinensis*, *T. lixianensis* vs. *T. orientalis*, and *Channa argus* vs. *Channa maculata*). In these cases, manual review was conducted to select the best option, using a combination of sample site information, known species occurrence data, and expert knowledge. Specifically,

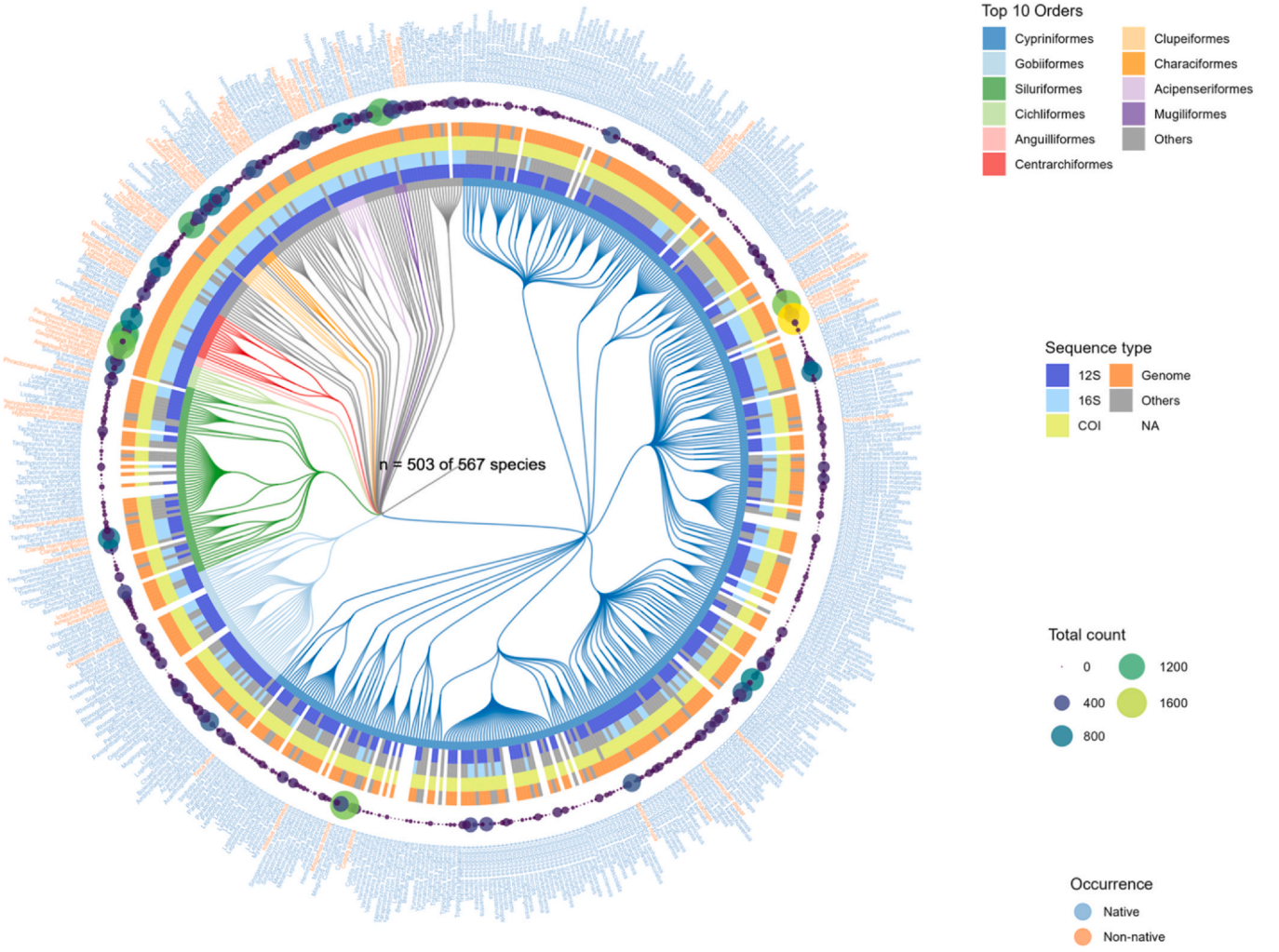



**Fig. 3.** Linean tree of the compiled list of fish species ($n$ = 567) in the Yangtze River Basin, highlighting the species with curated mitochondrial DNA sequences in YZFishDB ($n$ = 503). The branches and the innermost ring are colored by taxonomic order with the subsequent rings indicating the presence of the specified sequence types, e.g., 12S, 16S, COI. The outer points indicate the total number of sequences available. The color of the species name indicates native (blue; 430 of 467 curated) and non-native (red; 73 of 74 curated) species (see Table S3 for details).

comparative species distribution and occurrence were used to select the most likely taxa, with *S. sinensis* and *T. orientalis* selected over their counterparts. Due to the similarities in distribution, occurrence, and site locations, the ASVs with the ambiguous taxonomic assignments for *H. molitrix* vs. *H. nobilis*, *M. terminalis* vs. *P. pekinensis*, and *C. argus* and *C. maculata* were omitted from the final results.

In total, 134 unique species from 95 genera were identified by the YZFishDB database, and 125 unique species from 89 genera were identified by the MitoFish database, with both sets of taxa belonging to the same families and orders (Fig. 5). 118 species (83.7%) and 88 genera (91.67%) were shared in common between the two sets of results. Of the assigned species, YZFishDB identified 15 non-native species, while the MitoFish database identified one additional non-native species. Specifically, MitoFish identified *Oryzias latipes*, which YZFishDB identified as *Oryzias sinensis* based on the geographic distribution of both species. In contrast, the remaining non-native species identified were shared between both reference databases.

A total of 23 taxa were identified in only one of the reference databases (16 in YZFishDB; 7 in MitoFish; Fig. 5). All ASVs assigned to unique taxa by the MitoFish database were also identified by the YZFishDB database. In each of these cases, the differences in assigned taxa were due to the higher specificity of the BLAST alignment quality of additional reference sequences present in the YZFishDB database but absent in the MitoFish database (Table 1).

## 4. Discussion

We developed a curated DNA reference database for the fishes (Actinopterygii) of the Yangtze River Basin (YRB), consolidating reference sequences from four major public DNA reference databases (BOLD, MIDORI2, MitoFish, NCBI). The curation process during the development of our reference database confirmed that there was high cross-referencing between the databases, as expected given the nature of public data. For DNA reference sequences of our target taxonomic group, i.e., Actinopterygii, most relied on the NCBI nucleotide database as the repository, with some exceptions, e.g., certain COI sequences from the BOLD database. Due to the reliance on the NCBI database as the primary repository, a large number of exact duplicates (species-sequence duplicates) were found when first comparing the all retrieved sequences. However, the main issue after the initial basic data standardization and deduplication was sequences with multiple taxonomic matches, i.e., many-to-one species-sequence relationship. These ambiguous sequences can be largely categorized into three groups: (1) real instances of identical sections of DNA sequences shared amongst sister taxa, (2) mislabeled data due to non-standardized naming conventions (abbreviations, invalid taxon names, typos) or incorrect taxonomic identifications, and (3) outdated data retained in the databases.

Although there are some known sister taxa in the YRB with highly similar sections of DNA sequences, typically, these tend not to be exact

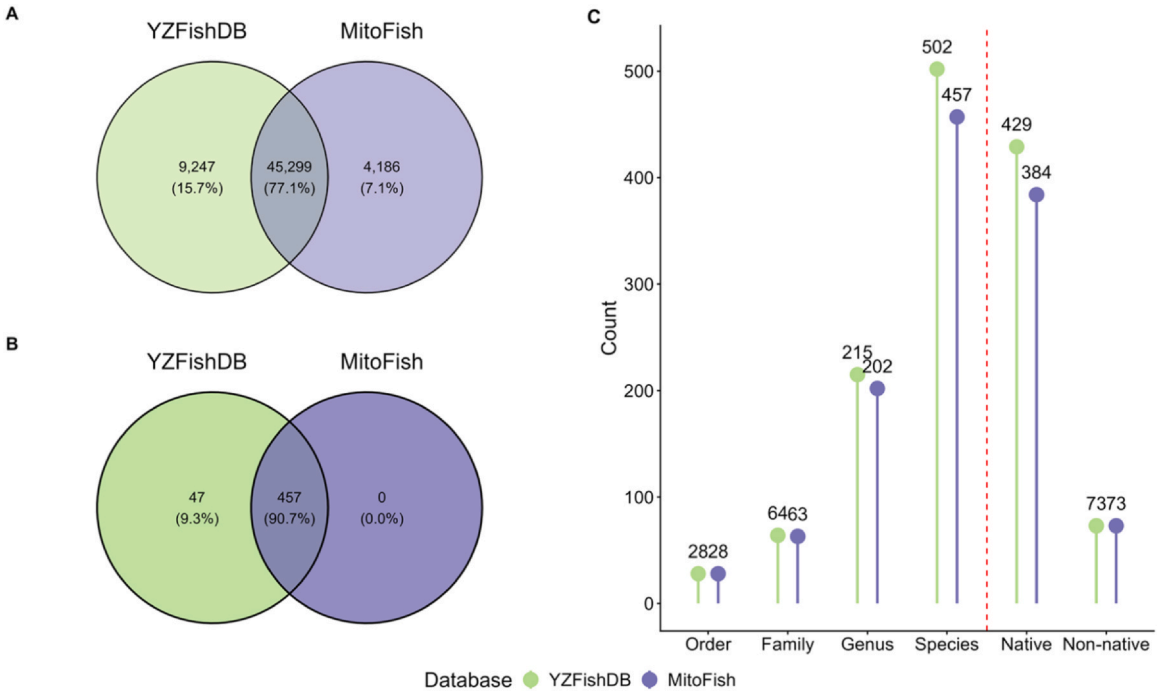



**Fig. 4.** Comparative assessment of YZFishDB and MitoFish reference databases. (**A**) reference sequence (ASV) coverage; (**B**) taxonomic overlap and (**C**) Taxonomic composition and native/non-native species representation.

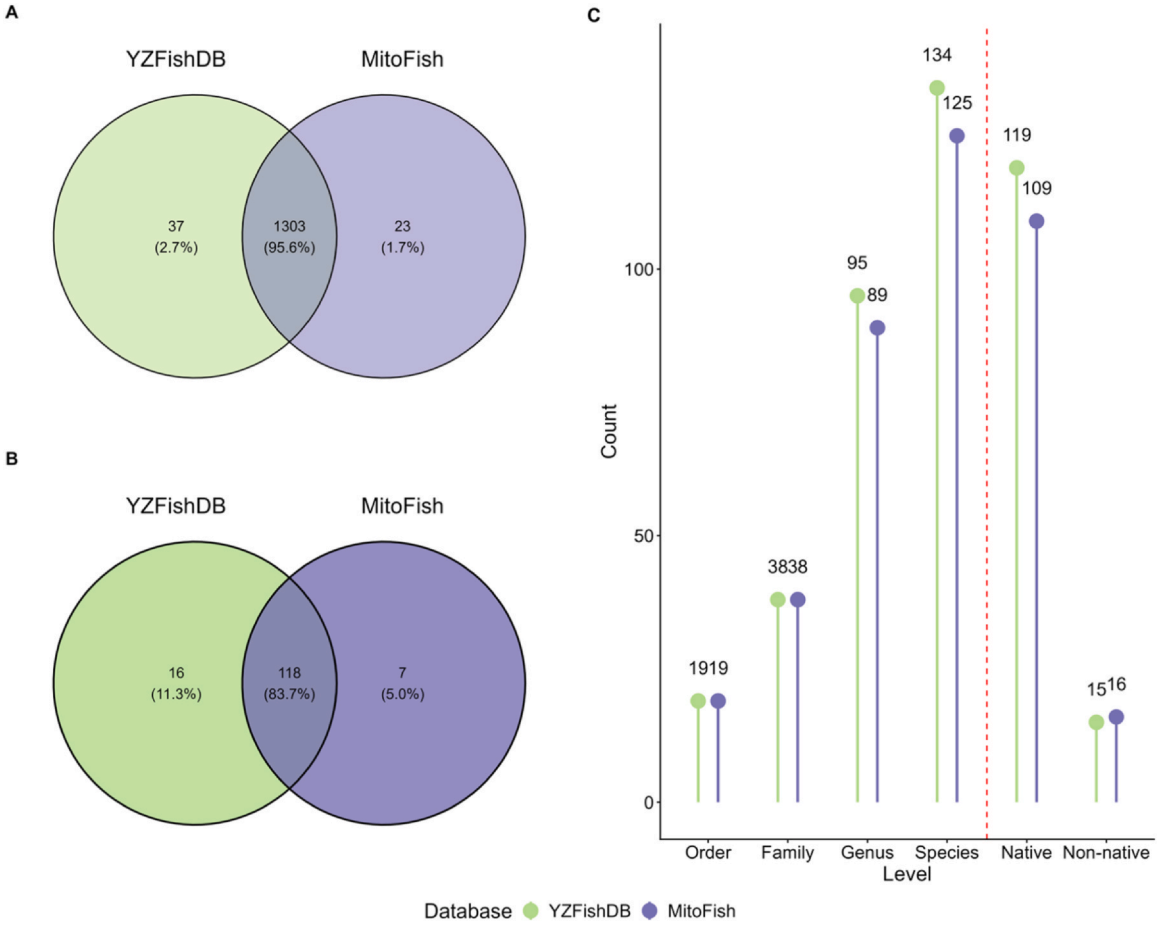



**Fig. 5.** Comparative assessment of YZFishDB and MitoFish reference databases based on taxonomic assignments from Yangtze River Basin eDNA data. (**A**) ASV assignment coverage; (**B**) Taxonomic overlap; and (**C**) Taxonomic composition and native/non-native species representation.

**Table 1**
Comparisons of key taxonomic assignment quality metrics of amplicon sequence variants (ASVs) assignments between YZFishDB and MitoFish for the ASVs that were assigned to unique taxa by the MitoFish reference database. "Count" indicates the total number of hits by the associated taxa; "bitscore_pb" indicates the maximum bitscore per base; "pident" indicates the maximum percent identity; "evalue" indicates minimum e-value; "ccovs" indicates maximum combined coverage. Assignment with better quality metrics are highlighted in bold.

| | RefDB | ASV | Species | Count | bitscore_pb | pident | evalue | ccovs |
|---|---|---|---|---|---|---|---|---|
| **1** | **YZFishDB** | **ASV_00746** | ***Megalobrama pellegrini*** | **5** | **1.80** | **99.11** | $\mathbf{1.22\times10^{-112}}$ | **46.41** |
| 2 | MitoFish | ASV_00746 | *Megalobrama skolkovii* | 2 | 1.80 | 99.11 | $1.10\times10^{-112}$ | 11.16 |
| **3** | **YZFishDB** | **ASV_00929** | ***Megalobrama pellegrini*** | **5** | **1.80** | **99.11** | $\mathbf{1.22\times10^{-112}}$ | **46.41** |
| 4 | MitoFish | ASV_00929 | *Megalobrama skolkovii* | 2 | 1.80 | 99.11 | $1.10\times10^{-112}$ | 11.16 |
| **5** | **YZFishDB** | **ASV_00967** | ***Megalobrama pellegrini*** | **5** | **1.80** | **99.11** | $\mathbf{1.22\times10^{-112}}$ | **46.41** |
| 6 | MitoFish | ASV_00967 | *Megalobrama skolkovii* | 2 | 1.80 | 99.11 | $1.10\times10^{-112}$ | 11.16 |
| **7** | **YZFishDB** | **ASV_00995** | ***Siniperca chuatsi*** | **2** | **1.80** | **99.06** | $\mathbf{1.59\times10^{-106}}$ | **44.62** |
| 8 | MitoFish | ASV_00995 | *Siniperca loona* | 2 | 1.80 | 99.06 | $1.44\times10^{-106}$ | 10.69 |
| **9** | **YZFishDB** | **ASV_01105** | ***Siniperca chuatsi*** | **2** | **1.80** | **99.06** | $\mathbf{1.59\times10^{-106}}$ | **44.62** |
| 10 | MitoFish | ASV_01105 | *Siniperca loona* | 2 | 1.80 | 99.06 | $1.44\times10^{-106}$ | 10.69 |
| **11** | **YZFishDB** | **ASV_01298** | ***Siniperca chuatsi*** | **2** | **1.80** | **99.06** | $\mathbf{1.59\times10^{-106}}$ | **44.62** |
| 12 | MitoFish | ASV_01298 | *Siniperca loona* | 2 | 1.80 | 99.06 | $1.44\times10^{-106}$ | 10.69 |
| **13** | **YZFishDB** | **ASV_01724** | ***Siniperca chuatsi*** | **2** | **1.82** | **99.53** | $\mathbf{3.41\times10^{-108}}$ | **44.62** |
| 14 | MitoFish | ASV_01724 | *Siniperca loona* | 2 | 1.82 | 99.53 | $3.09\times10^{-108}$ | 10.69 |
| **15** | **YZFishDB** | **ASV_01744** | ***Siniperca chuatsi*** | **2** | **1.80** | **99.07** | $\mathbf{4.41\times10^{-107}}$ | **44.72** |
| 16 | MitoFish | ASV_01744 | *Siniperca loona* | 2 | 1.80 | 99.07 | $4.00\times10^{-107}$ | 10.72 |
| **17** | **YZFishDB** | **ASV_01753** | ***Siniperca chuatsi*** | **2** | **1.82** | **99.53** | $\mathbf{3.41\times10^{-108}}$ | **44.62** |
| 18 | MitoFish | ASV_01753 | *Siniperca loona* | 2 | 1.82 | 99.53 | $3.09\times10^{-108}$ | 10.69 |
| **19** | **YZFishDB** | **ASV_02177** | ***Siniperca chuatsi*** | **2** | **1.82** | **99.53** | $\mathbf{3.41\times10^{-108}}$ | **44.62** |
| 20 | MitoFish | ASV_02177 | *Siniperca loona* | 2 | 1.82 | 99.53 | $3.09\times10^{-108}$ | 10.69 |
| **21** | **YZFishDB** | **ASV_03688** | ***Acheilognathus macropterus*** | **3** | **1.82** | **99.44** | $\mathbf{1.57\times10^{-111}}$ | **85.44** |
| 22 | MitoFish | ASV_03688 | *Acheilognathus barbatus* | 1 | 1.80 | 99.10 | $1.43\times10^{-111}$ | 11.00 |
| **23** | **YZFishDB** | **ASV_05043** | ***Acheilognathus macropterus*** | **3** | **1.85** | **100.00** | $\mathbf{5.66\times10^{-111}}$ | **85.44** |
| 24 | MitoFish | ASV_05043 | *Acheilognathus barbatus* | 1 | 1.80 | 99.10 | $5.13\times10^{-111}$ | 10.98 |
| **25** | **YZFishDB** | **ASV_06224** | ***Acheilognathus macropterus*** | **3** | **1.82** | **99.51** | $\mathbf{1.57\times10^{-111}}$ | **85.44** |
| 26 | MitoFish | ASV_06224 | *Acheilognathus barbatus* | 1 | 1.80 | 99.10 | $1.43\times10^{-111}$ | 11.00 |
| **27** | **YZFishDB** | **ASV_06863** | ***Spinibarbus caldwelli*** | **3** | **1.80** | **99.09** | $\mathbf{2.64\times10^{-109}}$ | **45.38** |
| 28 | MitoFish | ASV_06863 | *Spinibarbus hollandi* | 1 | 1.80 | 99.09 | $2.39\times10^{-109}$ | 10.92 |
| **29** | **YZFishDB** | **ASV_07559** | ***Schizopygopsis eckloni*** | **6** | **1.82** | **99.55** | $\mathbf{4.38\times10^{-112}}$ | **45.87** |
| 30 | MitoFish | ASV_07559 | *Gymnodiptychus pachycheilus* | 1 | 1.82 | 99.55 | $3.97\times10^{-112}$ | 11.07 |
| **31** | **YZFishDB** | **ASV_07672** | ***Spinibarbus caldwelli*** | **5** | **1.82** | **99.54** | $\mathbf{5.66\times10^{-111}}$ | **45.38** |
| 32 | MitoFish | ASV_07672 | *Spinibarbus hollandi* | 1 | 1.82 | 99.54 | $5.13\times10^{-111}$ | 10.92 |
| **33** | **YZFishDB** | **ASV_07673** | ***Spinibarbus caldwelli*** | **5** | **1.82** | **99.54** | $\mathbf{5.66\times10^{-111}}$ | **45.38** |
| 34 | MitoFish | ASV_07673 | *Spinibarbus hollandi* | 1 | 1.82 | 99.54 | $5.13\times10^{-111}$ | 10.92 |
| **35** | **YZFishDB** | **ASV_07866** | ***Spinibarbus caldwelli*** | **5** | **1.82** | **99.54** | $\mathbf{5.66\times10^{-111}}$ | **45.38** |
| 36 | MitoFish | ASV_07866 | *Spinibarbus hollandi* | 1 | 1.82 | 99.54 | $5.13\times10^{-111}$ | 10.92 |
| **37** | **YZFishDB** | **ASV_08335** | ***Schizopygopsis kialingensis*** | **2** | **1.80** | **99.10** | $\mathbf{2.04\times10^{-110}}$ | **45.82** |
| 38 | MitoFish | ASV_08335 | *Gymnodiptychus pachycheilus* | 1 | 1.80 | 99.10 | $1.85\times10^{-110}$ | 11.07 |
| **39** | **YZFishDB** | **ASV_08664** | ***Spinibarbus caldwelli*** | **3** | **1.80** | **99.09** | $\mathbf{2.64\times10^{-109}}$ | **45.38** |
| 40 | MitoFish | ASV_08664 | *Spinibarbus hollandi* | 1 | 1.80 | 99.09 | $2.39\times10^{-109}$ | 10.92 |
| **41** | **YZFishDB** | **ASV_08875** | ***Acheilognathus macropterus*** | **3** | **1.85** | **100.00** | $\mathbf{5.66\times10^{-111}}$ | **85.44** |
| 42 | MitoFish | ASV_08875 | *Acheilognathus barbatus* | 1 | 1.80 | 99.10 | $5.13\times10^{-111}$ | 10.98 |
| **43** | **YZFishDB** | **ASV_09570** | ***Acheilognathus macropterus*** | **3** | **1.85** | **100.00** | $\mathbf{1.22\times10^{-112}}$ | **85.44** |
| 44 | MitoFish | ASV_09570 | *Acheilognathus barbatus* | 1 | 1.82 | 99.55 | $1.10\times10^{-112}$ | 10.98 |
| **45** | **YZFishDB** | **ASV_09885** | ***Schizopygopsis eckloni*** | **6** | **1.82** | **99.55** | $\mathbf{4.38\times10^{-112}}$ | **45.87** |
| 46 | MitoFish | ASV_09885 | *Gymnodiptychus pachycheilus* | 1 | 1.82 | 99.55 | $3.97\times10^{-112}$ | 11.07 |
| **47** | **YZFishDB** | **ASV_11877** | ***Acheilognathus macropterus*** | **5** | **1.85** | **100.00** | $\mathbf{2.62\times10^{-114}}$ | **85.44** |
| 48 | MitoFish | ASV_11877 | *Acheilognathus barbatus* | 1 | 1.85 | 100.00 | $2.37\times10^{-114}$ | 10.98 |
| **49** | **YZFishDB** | **ASV_12341** | ***Oryzias sinensis*** | **2** | **1.80** | **99.06** | $\mathbf{5.70\times10^{-106}}$ | **44.35** |
| 50 | MitoFish | ASV_12341 | *Oryzias latipes* | 1 | 1.80 | 99.06 | $5.17\times10^{-106}$ | 10.70 |
| **51** | **YZFishDB** | **ASV_22689** | ***Acheilognathus macropterus*** | **3** | **1.82** | **99.55** | $\mathbf{1.22\times10^{-112}}$ | **85.44** |
| 52 | MitoFish | ASV_22689 | *Acheilognathus barbatus* | 1 | 1.82 | 99.55 | $1.10\times10^{-112}$ | 10.98 |
| **53** | **YZFishDB** | **ASV_29615** | ***Claea dabryi*** | **2** | **1.85** | **100.00** | $\mathbf{1.57\times10^{-111}}$ | **45.09** |
| 54 | MitoFish | ASV_29615 | *Triplophysa bleekeri* | 1 | 1.85 | 100.00 | $1.43\times10^{-111}$ | 10.80 |
| **55** | **YZFishDB** | **ASV_40420** | ***Claea dabryi*** | **2** | **1.80** | **99.08** | $\mathbf{3.41\times10^{-108}}$ | **45.09** |
| 56 | MitoFish | ASV_40420 | *Triplophysa bleekeri* | 1 | 1.80 | 99.08 | $3.09\times10^{-108}$ | 10.80 |

duplicates. Therefore, these similar sequences were retained in our database to ensure inclusivity and comprehensiveness. However, in real-world applications, such as eDNA samples, the length of the amplified DNA metabarcoding fragments is often imperfect and tends to be shorter. This leads to the emergence of taxonomic assignment ambiguities, where multiple taxa may have identical assignment quality metrics to a single ASV. In these instances, selecting the appropriate taxa would require manual verification based on the relevant taxa's geographical range and ecology. In our study, taxonomic assignments using YZFishDB and MitoFish both identified multiple best matches, with hits sharing identical scores across all metrics. These ambiguous matches accounted for a small fraction of all taxonomic assignments in YZFishDB and MitoFish (0.8% and 1.2%, respectively), with minimal impact on the final results.

In addition to having fewer ambiguous matches using YZFishDB, we found that the both sets of results identified ambiguous matches between *Triplophysa orientalis* and *Triplophysa lixianensis*, species that have been found to exhibit high intraspecific and low interspecific K2P distances (Li et al., 2017). While we were able to discerned the most likely species based on known geographical occurrences and the sample site, other ambiguous matches, particularly those found in the MitoFish taxonomic assignment were resolved through our database curation

processes. For example, the ambiguous matches between *Megalobrama terminalis* and *Parabramis pekinensis* were absent in the taxonomic assignments using YZFishDB, which indicated potential sequence mislabeling or misidentifications.

Our database curation process targeted two aspects to overcome this issue, i.e., through nomenclature standardization and sequence divergence checks. To this end, we automated standardization for most mislabeled sequences resulting from non-standardized naming conventions by relying on authoritative sources for fishes, e.g., Eschmeyer's Catalog of Fishes (Fricke et al., 2025). Even though this process may have overlooked some exceptions, particularly data with unusual typos or highly unconventional metadata records, the use of a curated taxonomic lookup database streamlined the entire curation workflow. It ensured that the nomenclature standardization was not only reproducible but also easily improvable and updatable.

Sequences with incorrect taxonomic identification, as well as outdated data in public databases, accounted for the majority of the ambiguous sequences identified during our data curation process. During the manual curation of the ambiguous sequences, i.e., identical sequences attributed to different taxa, more than 94% (610 of 647 sequences) were traced back to their sources but remained indistinguishable (Table S6). Many of these sources had complete metadata for different species, often within the same database, where these ambiguous sequences were associated with taxa that were difficult to distinguish based on morphological traits or were associated with hybrids. Although it is possible that some of these entries are outdated and may have been real misidentifications, it is difficult to assess the status of these entries on public databases that serve as repositories and archives, particularly when erroneous entries remain in place for long periods (Keck et al., 2023; Magoga et al., 2022).

Most of the ambiguous sequences we curated were attributed to taxa that were already relatively well-represented and, as such, were removed from the database to maintain database quality (see "Data processing and quality control"). To further ensure that mislabeled data, as highlighted by our ambiguous sequence curation process, would not affect the overall quality of our database, we included a divergence check based on the phylogenetic relationship between sequences from the same gene and species, filtering out remaining sequences that were notably different from their cohort. While our post hoc barcode gap analysis indicated that some sequences with high intraspecific distance remain in the database, we found that the majority of sequence-marker groups showed clear median barcode gaps.

While other public databases, e.g., BOLD, were included in our database development, our choice to only compare the taxonomic assignments with MitoFish was due to sample data constraints (only 12S sequences) and the popularity of MitoFish as one of the primary reference database for many studies in this region (Deng et al., 2024; Geng et al., 2025; Shen et al., 2024; Zhang et al., 2019). Although we expected all species identified in the MitoFish reference database to be present in the YZFishDB taxonomic assignment results, five taxa were unique to the MitoFish assignment. Upon closer inspection, the corresponding ASVs of these unique MitoFish taxonomic assignments were also assigned to taxa that had better assignment quality metrics using YZFishDB, whereby these differences occurred mainly due to the poorer coverage of MitoFish in comparison to YZFishDB. In addition to wider taxonomic coverage from YZFishDB, the improved sequence specificity also helped improve taxonomic assignment confidence, which was able to help break ties for sequences sharing similar qualities. Despite the high proportion of taxonomic overlap between YZFishDB and MitoFish, these differences further illustrate the importance of well-curated reference databases, where the improvements in specificity and coverage can affect subsequent results.

By optimizing our QC pipeline, we were able to effectively flag serious issues with the relevant sequences, e.g., contaminant, numts, and divergence. Nonetheless, a common aspect across all these issues we encountered during the development of our DNA reference database was the need for manual curation. The selection of ambiguous sequences, both during the standard deduplication process and in the final selection of taxonomic assignments, and the compilation of our taxonomic list for the freshwater fishes of the YRB highlighted the crucial role that manual verification and expert knowledge played throughout the development process. When compiling our taxonomic list, we followed the principle of inclusivity, including species mentioned as present by any source. However, for some groups, like the genus *Megalobrama*, species nomenclature remains vague as multiple names continue to be applied to describe different species within the genus (Bai et al., 2015; Chen and Wang, 2021; Dong et al., 2022; Liu et al., 2024; Yin et al., 2024). In these exceptional cases, we relied on both authoritative sources, i.e., Eschmeyer's Catalog of Fishes and expert knowledge to help standardize the nomenclature.

As the basin-wide fishing ban in the YRB has past its halfway mark (Liu et al., 2023), eDNA surveys continues to increase in popularity in place of traditional fisheries sampling methods particularly in this region. Although the combination of traditional fisheries sampling and eDNA sampling is considered the gold standard for biodiversity surveys, particularly for fish assemblages, the reduced access to traditional fisheries methods has indirectly encouraged the rapid adoption of eDNA sampling as the primary technique in the YRB. As mentioned earlier, one shared issue of these studies is the reliance on public reference databases, which requires additional post hoc filtering for the taxonomic assignment results (Qian et al., 2023) to avoid unwanted assignments (e.g., assignments to non-targeted organisms), false positives due to low quality sequences (e.g., short references or misattributed references), and missing data for rare species.

By using a curated database like YZFishDB, which is specific to the study region, we can avoid potential subjective post hoc selection of taxonomic assignments and have greater confidence in the taxonomic assignments than with uncurated reference databases. False positives are much more consequential in the context of invasion ecology where the presence of genetically similar taxa within unspecified reference databases can potentially lead to convincing misidentifications when relying solely on assignment quality metrics (Fonseca et al., 2023). Therefore, surveys aimed at invasive species monitoring not only require reference databases containing sequences of species with known occurrence in the region but also include sequences of suspected non-native species. By including potential non-native species and several novel sequences for underrepresented fish species, YZFishDB not only expands taxonomic coverage but also provides curated screening for potential non-native species.

The comparison of ASV assignments between the two reference databases for the unique taxa identified by MitoFish showed that the assignments by YZFishDB were supported by more sequence matches and generally higher assignment quality metrics (Table 1). Of the unique taxa that were assigned by YZFishDB, two taxa (*Discogobio laticeps* and *Sinocrossocheilus labiatus*) were from the novel sequences added from our local samples, which points to the importance of increasing the taxonomic coverage of reference databases with data from rarer taxa. As both reference databases were filtered based on our YRB fish list, most of the non-native taxa identified were identical between the two. The only difference was *Oryzias latipes*, a non-native species in the YRB, which was likely a misassignment from the MitoFish database due to differences in sequence coverage (ASV 12341; Table 1).

Although the comparison between YZFishDB and MitoFish suggests that taxonomic assignments based on YZFishDB tend to have better assignment quality metrics compared to MitoFish, we acknowledge that this comparison does not fully validate the accuracy of YZFishDB, which is the major limitation of this study. Ideally, validation of such reference databases would be conducted using samples with known fish assemblages, mock communities, or by comparing them to concurrent traditional fisheries samples, which we unfortunately do not have access to. Although YZFishDB can be considered as a curated reference database, it may still contain some erroneous sequences stemming from

upstream errors as suggested by our post hoc barcode gap analysis.

In the context of non-native species detection, even though YZFishDB included sequences from several potential species, we recommend that users further supplement the reference database with sequences from a targeted list of species that may be absent from our species list. While we demonstrated the taxonomic assignment results of YZFishDB using BLASTN with the database as is, additional bioinformatics steps would be required to prepare the reference database for use with more stringent taxonomic assignment methods, e.g., DADA2.

## 5. Conclusion

In conclusion, by developing and sharing a well-curated database for important ecosystems like the YRB for a key assemblage (freshwater fishes), we hope to contribute to improving the accuracy, standards, and data accessibility of biodiversity monitoring and ecological research in the YRB.

## Funding

Funding for the research was provided by the National Key R&D Program of China (2024YFF1307500), the National Natural Science Foundation of China (32470551), the research project “Changes in aquatic biological resources and diversity after the fishing ban and protection policies after the ban period” (JGGJ0425040, JGGJ0425041), the Key R&D Program of Jiangxi Province (2023BBG70011), and the Chinese Academy of Sciences President's International Fellowship Initiative (2026PD0172). This work was also supported by the Chinese Academy of Sciences-Alliance of International Science Organization (CAS-ANSO) Scholarship for Young Talents.

## CRediT authorship contribution statement

**Zi Xun Kua:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Houhua Bing:** Resources, Investigation. **Yue Wu:** Writing – review & editing. **Yanjun Shen:** Writing – review & editing. **Sebastien Brosse:** Writing – review & editing. **Jing Zhang:** Writing – review & editing, Resources. **Yuxin Hu:** Writing – review & editing, Resources. **Jiayang He:** Writing – review & editing. **Jun Xu:** Writing – review & editing, Funding acquisition. **Guohuan Su:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

## Data availability

All data and scripts relating to the database curation and subsequent taxonomic assignment comparison are available on Zenodo (https://doi.org/10.5281/zenodo.18155084).

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Guohuan Su is an editorial board member for Aquatic Diversity and Ecology and was not involved in the editorial review or the decision to publish this article.

## Acknowledgement

We would like to thank the various field crews that helped collect and prepared the field samples. We are particularly grateful to Professor Shunping He for his insightful comments, constructive suggestions, and professional guidance, which greatly improved this work.

## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ade.2026.100009.

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